

# Companion to Bird & de Moor, Algebra of Programming

## §8. Thinning Algorithms

### §8.1. Thinning

(8.1)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$  (8.1).

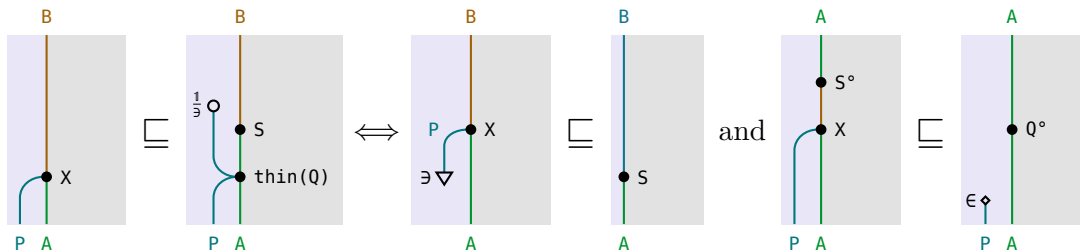
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q^\circ \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(8.1a)

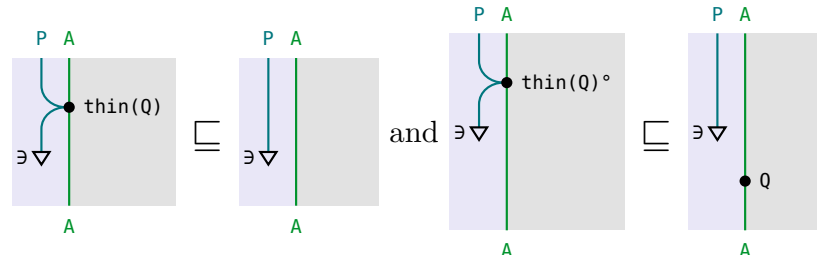
| the law                                                                                                                                                                                                                                                                                            | what it says                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \exists S$ and $S^\circ X \sqsubseteq Q^\circ \epsilon$                                                                                                                                                                             | everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones                             |
| $\text{thin}(Q) \exists \sqsubseteq \exists$                                                                                                                                                                                                                                                       | everything a thinning keeps was in the set                                                                           |
| $\text{thin}(Q)^\circ \exists \sqsubseteq \exists Q$                                                                                                                                                                                                                                               | every element of the set has a Q-lower bound among the kept ones                                                     |
| $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$                                                                                                                                                                                                                               | the fewer pairs Q relates, the fewer subsets count as thinnings                                                      |
| $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$<br>$Q \ Q \sqsubseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \sqsubseteq \text{thin}(Q)$                                                                                                                                               | keeping everything is always a legal thinning                                                                        |
| $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$<br>$Q \sqsubseteq R, 1 \sqsubseteq Q, R R \sqsubseteq R$ — (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \sqsubseteq R$ follows | <b>thin-introduction:</b> thinning first cannot lose an R-minimum                                                    |
| $\text{thin}(Q) \sqsupseteq \text{est}(Q) \ \frac{1}{\exists}$ (8.2) — (8.2)                                                                                                                                                                                                                       | <b>thin-elimination:</b> keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on |
| $\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant                                                                                                                         | the usable variant: R need only refine Q between values S gives one argument                                         |
| $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$ (8.4)                                                                                                                                                                                                                           | thinning each member set is a thinning of the union                                                                  |

#### §8.1.1. $X \sqsubseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \exists S$ and $S^\circ X \sqsubseteq Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)



(8.1.1a) at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$ , since  $\exists \exists \exists = \Lambda(\exists) = 1$

#### §8.1.2. $\text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ given $Q \sqsubseteq R, 1 \sqsubseteq Q, R R \sqsubseteq R$

**$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$**  (8.1.2a)

the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$

(a) (b)

|     |                                 |                                         |
|-----|---------------------------------|-----------------------------------------|
| (a) | $\text{est}(R)$                 | $1 \sqsubseteq \text{thin}(Q)$ — (8.1a) |
| (b) | $\text{thin}(Q) \text{ est}(R)$ |                                         |

**$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$**  (8.1.2b)

the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangleq \text{thin}(Q) \text{ est}(R)$  —  $\text{est-up}$

(a) (b) (c)

|     |                                 |                                                         |
|-----|---------------------------------|---------------------------------------------------------|
| (a) | $\text{thin}(Q) \text{ est}(R)$ | $\text{est}(R) \sqsubseteq \exists$ — est-laws          |
| (b) | $\text{thin}(Q) \exists$        | $\text{thin}(Q) \exists \sqsubseteq \exists$ — (8.1.1a) |
| (c) | $\exists$                       |                                                         |

**$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$**  (8.1.2c)

the  $\sqsubseteq$  half, second condition —  $Q \sqsubseteq R$ ,  $R$  transitive

(a) (b) (c) (d) (e)

|     |                                     |                                                         |
|-----|-------------------------------------|---------------------------------------------------------|
| (a) | $\in \text{thin}(Q) \text{ est}(R)$ | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (8.1.1a) |
| (b) | $Q^\circ \in \text{est}(R)$         | $\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws      |
| (c) | $Q^\circ R^\circ$                   | $Q \sqsubseteq R$                                       |
| (d) | $R^\circ R^\circ$                   | $R$ transitive                                          |
| (e) | $R^\circ$                           |                                                         |

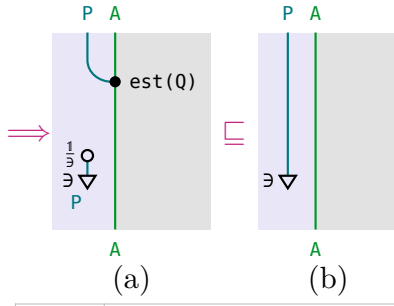
$$\text{est}(Q) \stackrel{1}{\exists} \sqsubseteq \text{thin}(Q)$$

(8.2)

the singleton holding a  $Q$ -least member of a set is a thinning of that set

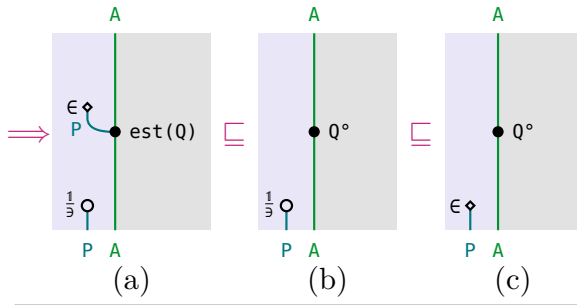
$$\text{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

every member of the singleton is a member of the set



|     |                                           |                                                                                        |
|-----|-------------------------------------------|----------------------------------------------------------------------------------------|
| (a) | $\text{est}(Q) \stackrel{1}{\exists} \ni$ | $\stackrel{1}{\exists} \ni = 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \ni$ — est-laws |
| (b) | $\ni$                                     | $\triangleq \text{thin}, \epsilon \setminus (Q^\circ \epsilon)$ half                   |

$$\epsilon \text{ est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in  $Q$  to every member of the set

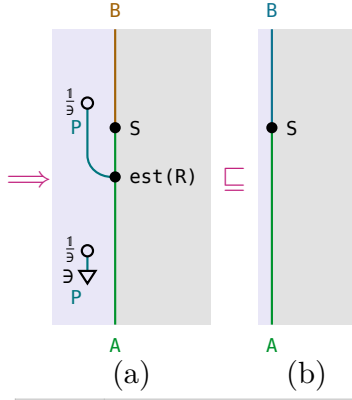
|     |                                                 |                                                         |
|-----|-------------------------------------------------|---------------------------------------------------------|
| (a) | $\epsilon \text{ est}(Q) \stackrel{1}{\exists}$ | $\epsilon \text{ est}(Q) \sqsubseteq Q^\circ$ — est-up  |
| (b) | $Q^\circ \stackrel{1}{\exists}$                 | $\stackrel{1}{\exists} \sqsubseteq \epsilon$ — pow-laws |
| (c) | $Q^\circ \epsilon$                              |                                                         |

$$R \cap S^\circ S \sqsubseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q) \quad (8.3)$$

given  $Rn(S^\circ S) \sqsubseteq Q$ ,  $Q$  a preorder

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni \sqsubseteq S$$

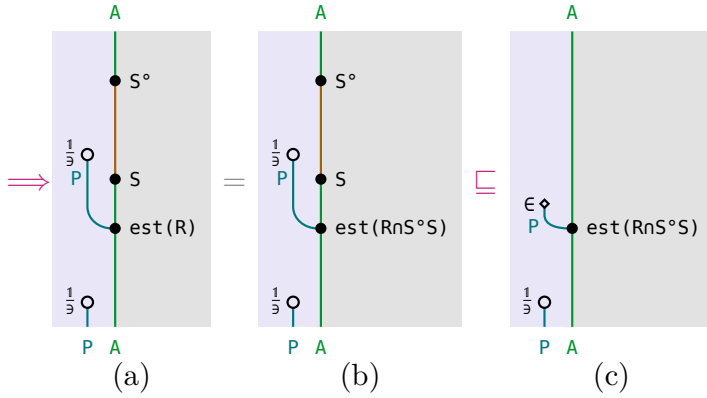
every member of the singleton is a value  $S$  returns



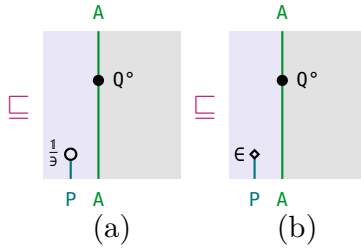
|     |                                             |                                                                                      |
|-----|---------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $\frac{S}{S} \text{est}(R) \frac{1}{S} \ni$ | $\frac{1}{S} \ni 1$ — pow-laws; $\frac{S}{S} \text{est}(R) \sqsubseteq S$ — est-laws |
| (b) | $S$                                         | (8.1a), second condition                                                             |

$$R \cap S^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument



|     |                                                                |                                                                                                       |
|-----|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| (a) | $S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S}$                | $\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(RnS^\circ S)$ — est-laws                          |
| (b) | $S^\circ \frac{S}{S} \text{est}(R \cap S^\circ S) \frac{1}{S}$ | $S^\circ \frac{S}{S} \sqsubseteq \in$ — pow-laws                                                      |
| (c) | $\in \text{est}(R \cap S^\circ S) \frac{1}{S}$                 | $\in \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$ — est-up; $Rn(S^\circ S) \sqsubseteq Q$ |



|     |                       |                                          |
|-----|-----------------------|------------------------------------------|
| (a) | $Q^\circ \frac{1}{S}$ | $\frac{1}{S} \sqsubseteq \in$ — pow-laws |
| (b) | $Q^\circ \in$         |                                          |

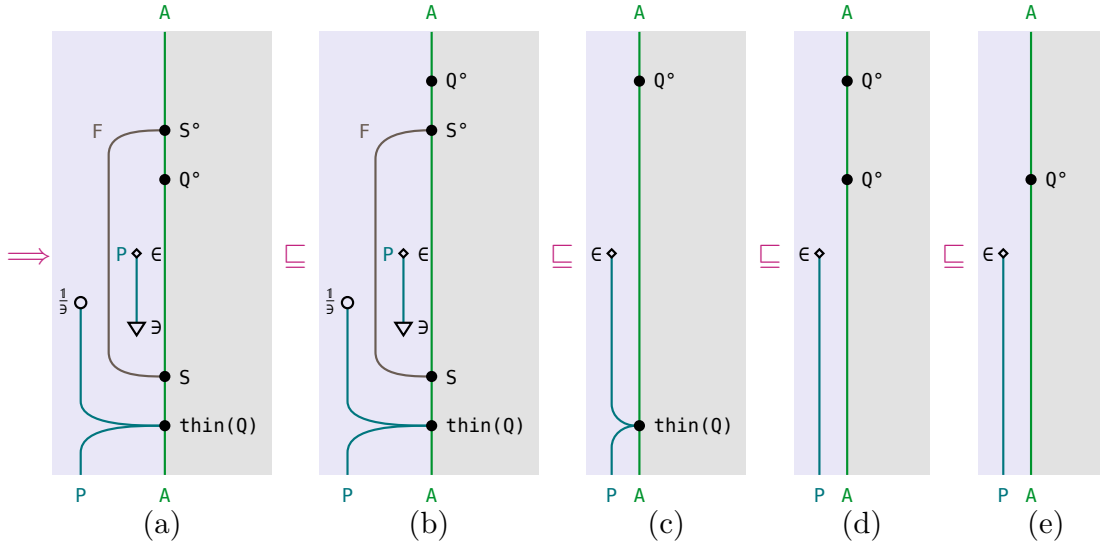
$$Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \subseteq \frac{(S)}{\exists} \text{thin}(Q)$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set —  $S$  monotonic on  $Q$ ,  $Q$  a preorder

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \supseteq (S) \text{ and } (S) \circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \subseteq Q^\circ \in$$

(8.1a) at  $X \triangleq (\frac{F(\exists) S}{\exists} \text{thin}(Q))$ ,  $(S)$  for its  $S$

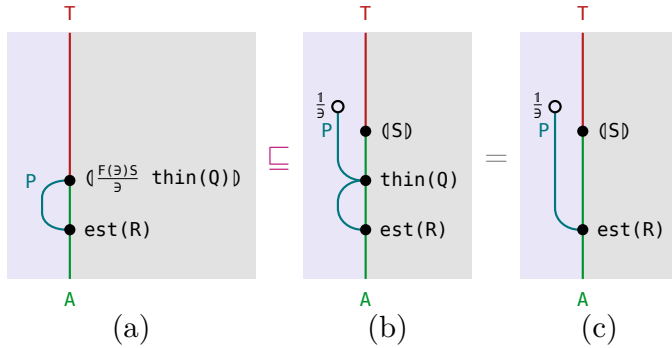


|     |                                                                      |                                                                         |
|-----|----------------------------------------------------------------------|-------------------------------------------------------------------------|
| (a) | $S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $S^\circ F(Q^\circ) \subseteq Q^\circ S^\circ$ — mon-str, relator-laws  |
| (b) | $Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $S^\circ F(\in) \frac{F(\exists) S}{\exists} \subseteq \in$ — pow-laws  |
| (c) | $Q^\circ \in \text{thin}(Q)$                                         | $\in \text{thin}(Q) \subseteq Q^\circ \in$ — $\triangleq$ thin, adj-all |
| (d) | $Q^\circ Q^\circ \in$                                                | $Q$ a preorder                                                          |
| (e) | $Q^\circ \in$                                                        |                                                                         |

$$Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{ est}(R) \subseteq \frac{(S)}{\exists} \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself —  $S$  monotonic on  $Q$ ,  $Q \subseteq R$ , both preorders



|     |                                                                |                                                          |
|-----|----------------------------------------------------------------|----------------------------------------------------------|
| (a) | $(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{ est}(R)$ | thinning theorem                                         |
| (b) | $\frac{(S)}{\exists} \text{thin}(Q) \text{ est}(R)$            | $\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — (8.1a) |
| (c) | $\frac{(S)}{\exists} \text{est}(R)$                            |                                                          |

## §8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$ ,  $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$ .

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$ ,  $([\text{wrapz}, \text{consw}]) = (1, \text{cost})$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$ .

(8.2b)

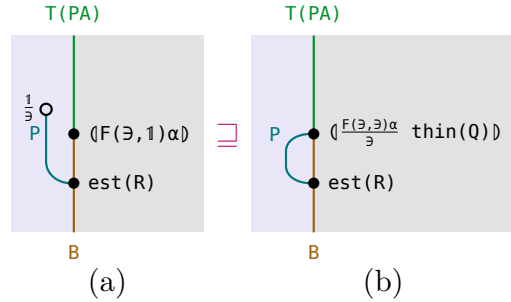
| the law                                                                       | what it says                                                                                                                                                           |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                           |
| $S \text{ head} \subseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \subseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

$$Q \subseteq R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R^\circ \quad R^\circ \subseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge R \cap (F(1, \exists) \alpha)^\circ \quad F(1, \exists) \alpha \subseteq Q \Rightarrow \quad (8.2c)$$

$$\lfloor \frac{F(\exists, 1)}{\exists} \rfloor P(\frac{F(1, \exists)}{\exists} \alpha \text{ est}(R)) \rfloor \text{est}(R) \subseteq \lfloor \frac{F(\exists, 1)}{\exists} \rfloor \alpha \rfloor \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

$$= \lfloor \frac{L(\exists)}{\exists} \rfloor \text{est}(R)$$



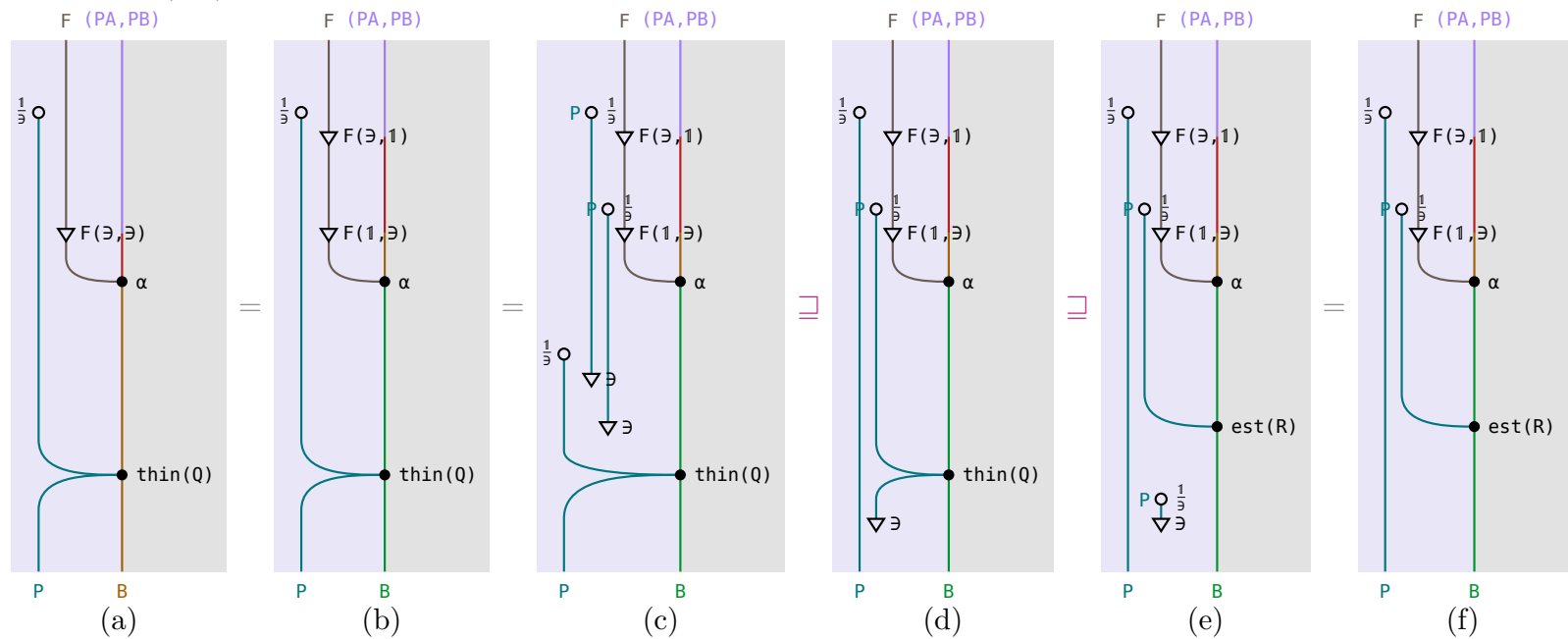
|     |                                                                                                           |                       |
|-----|-----------------------------------------------------------------------------------------------------------|-----------------------|
| (a) | $\lfloor \frac{F(\exists, 1)}{\exists} \rfloor \alpha \rfloor \text{est}(R)$                              | Corollary 8.1, (8.2b) |
| (b) | $\lfloor \frac{F(\exists, \exists)}{\exists} \rfloor \alpha \rfloor \text{thin}(Q) \rfloor \text{est}(R)$ |                       |

$\sqsubseteq$  (8.2d) under  $\lfloor \rfloor$  monotonic: the whole chain runs inside the reduce, and the  $\text{est}(R)$  behind it never moves.

$$R \cap (F(1, \exists) \alpha)^\circ \quad F(1, \exists) \alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} \alpha \text{ est}(R)) \subseteq \frac{F(\exists, \exists)}{\exists} \alpha \text{ thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

the algebra of (8.2c) row 2



|     |                                                                                                                         |                                                                                                                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists, \exists)}{\exists} \alpha \text{ thin}(Q)$                                                            | $F(\exists, \exists) = F(\exists, 1)F(1, \exists)$                                                                                                                       |
| (b) | $\frac{F(\exists, 1)}{\exists} \frac{F(1, \exists)}{\exists} \alpha \text{ thin}(Q)$                                    | $\frac{RS}{\exists} = \frac{R}{\exists} \frac{S}{\exists} \quad P(\frac{S}{\exists}) \text{ union, union} \triangle \frac{\exists \exists}{\exists}$                     |
| (c) | $\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} \alpha) \frac{\exists \exists}{\exists} \text{ thin}(Q)$ | $\text{union thin}(Q) \sqsupseteq P(\text{thin}(Q)) \text{ union} \quad (8.1a)$                                                                                          |
| (d) | $\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} \alpha \text{ thin}(Q)) \text{ union}$                   | $\frac{S}{\exists} \text{ thin}(Q) \sqsupseteq \frac{S}{\exists} \text{ est}(R) \quad \frac{1}{\exists} \text{ if } R \cap (S^\circ S) \subseteq Q \quad (8.1a), (8.2b)$ |
| (e) | $\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} \alpha \text{ est}(R) \frac{1}{\exists}) \text{ union}$  | $P(\frac{1}{\exists}) \text{ union} = 1$                                                                                                                                 |
| (f) | $\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R))$                               | $\frac{F(1, \exists)}{\exists} \alpha = \frac{F(1, \exists)}{\exists} P(\alpha)$                                                                                         |

$$= \text{cp}(F) P(\alpha) \text{ est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network  $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}]): \frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}] \triangle \text{cost}$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangle \text{cost}$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \times | \\ & & R & & R \quad S \\ & & \downarrow & & \downarrow \quad \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u  $F(R, S)$  v holds iff one of:

- u, v are both inl and  $d \ R \ d'$ : the leaf moves by R;
- u, v are both inr and  $p_1 \ R \ q_1 \wedge p_2 \ S \ q_2$ : the vertex moves by R, the rest by S.
- An inl never relates to an inr:  $R + S \triangle [R \text{ inl}, S \text{ inr}]$  returns to the summand it came from, and  $F(1) = 1$  allows no crossing.
- $F(\exists, \exists)$  relates  $\text{inr}(\{a, b\}, \{p, q\})$  to  $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$ , and  $\text{inl}(\{a, b\})$  to  $\text{inl}(a), \text{inl}(b)$ .

## §8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow PA$ ,  $\text{cup} : PA \times PA \rightarrow PA$ ,  $\text{cp}(F) \triangleq \text{subseq}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$  for  $\leq$  a connected preorder.

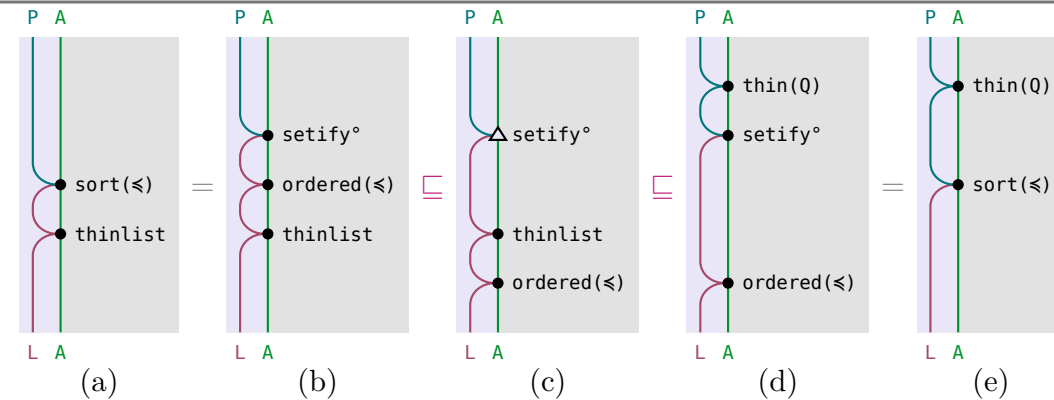
$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $Q[\text{nil}, \text{bump } Q]$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $\leq$  a connected preorder with both  $f_1, f_2$  monotonic on  $\leq$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

(8.3b)

| the law                                                                                                             | what it says                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\text{thinlist}(Q) \text{ } xs = [\text{minlist}(Q) \text{ } xs]$<br>(8.5), $Q$ connected, $xs$ non-empty          | what thinning should come to when it can: one element                                                              |
| $\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6)                 | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| $\text{sort}(\leq) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                              | a minimum of the sorted list is a minimum of the set                                                               |
| $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ (8.8)                             | shunt a function through a sort                                                                                    |
| $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), $p$ coreflexive                    | filtering a sorted list sorts the restricted set                                                                   |
| $(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup } \text{sort}(\leq)$ (8.10) | merging two sorted lists sorts their union                                                                         |
| $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$<br>(8.11), $F$ linear          | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

**Map(setify)  $\wedge$  Coreflexive(ordered( $\leq$ ))  $\wedge$  thinlist  $\sqsubseteq$  subseq  $\wedge$  ordered( $\leq$ ) subseq  $\sqsubseteq$  subseq** (8.6)  
**ordered( $\leq$ )  $\wedge$  thinlist setify  $\sqsubseteq$  setify thin( $Q$ )  $\Rightarrow$  sort( $\leq$ ) thinlist  $\sqsubseteq$  thin( $Q$ ) sort( $\leq$ )**  
 a thinning of the sorted list lists a thinning of the set —  $\leq$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .



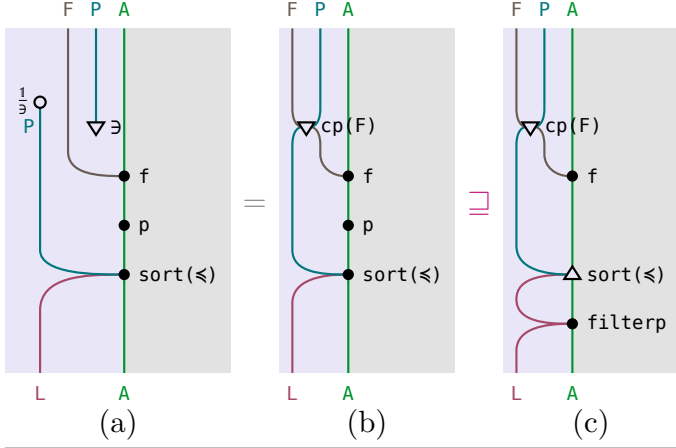
|     |                                                              |                                                                                                                                           |
|-----|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{sort}(\leq) \text{ thinlist}$                         | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ — } \triangleq \text{thinlist}$                            |
| (b) | $\text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$ | $\text{ordered}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q)$<br>$\text{ordered}(\leq) \text{ — } \triangleq \text{thinlist}$ |
| (c) | $\text{setify}^\circ \text{ thinlist ordered}(\leq)$         | $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq$<br>$\text{thinlist, dom-laws, triple-chains}$ |
| (d) | $\text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$  | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ — } \triangleq \text{thinlist}$                            |
| (e) | $\text{thin}(Q) \text{ sort}(\leq)$                          |                                                                                                                                           |

**Map(f)  $\wedge$  ( $\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$ )  $\wedge$  MonoAlg(f,  $\leq$ )  $\wedge$  sortF(f  $\leq$  f $^\circ$ ) listf  $\sqsubseteq$  P(f) sort( $\leq$ )  $\wedge$  sort( $\leq$ ) filterp  $\sqsubseteq$  E(p) sort( $\leq$ )  $\wedge$  F(sort( $\leq$ )) listcp  $\sqsubseteq$  cp(F) sortF(F( $\leq$ ))  $\Rightarrow$  F(sort( $\leq$ )) listcp listf filterp  $\sqsubseteq$   $\frac{F(\exists) f p}{\exists}$  sort( $\leq$ )** Lemma 8.1

one sorted list built from sorted arguments, instead of a set built and then sorted — **f** : **FA**→**A** monotonic on  $\leq$ , **p** coreflexive, **F** linear.

Map(f)  $\wedge$  sort( $\leq$ ) filterp  $\sqsubseteq$  E(p) sort( $\leq$ )  $\Rightarrow$  cp(F) P(f) sort( $\leq$ ) filterp  $\sqsubseteq$   $\frac{F(\exists) f p}{\exists}$  sort( $\leq$ )

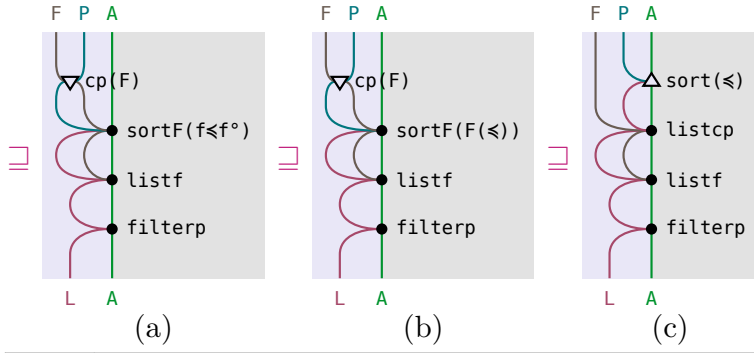
sorting the f-images of the product and filtering by p is below sorting the set F( $\exists$ )fp builds



|     |                                                     |                                                                                                                                             |
|-----|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$ | $\frac{F(\exists) fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist<br>$E(f)=P(f)$ , f a map — powrel-laws |
| (b) | cp(F) P(f) E(p) sort( $\leq$ )                      | sort( $\leq$ ) filter(p) $\sqsubseteq$ E(p) sort( $\leq$ ) — (8.3b)                                                                         |
| (c) | cp(F) P(f) sort( $\leq$ ) filterp                   | sort(f $\leq$ f $^\circ$ ) list(f) $\sqsubseteq$ P(f) sort( $\leq$ ) — (8.3b)                                                               |

Map(f)  $\wedge$  ( $\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$ )  $\wedge$  MonoAlg(f,  $\leq$ )  $\wedge$  sortF(f  $\leq$  f $^\circ$ ) listf  $\sqsubseteq$  P(f) sort( $\leq$ )  $\wedge$  F(sort( $\leq$ )) listcp  $\sqsubseteq$  cp(F) sortF(F( $\leq$ ))  $\Rightarrow$  F(sort( $\leq$ )) listcp listf filterp  $\sqsubseteq$  cp(F) P(f) sort( $\leq$ ) filterp

combining the arguments' sorted lists by listcp, mapping f and filtering is below sorting the f-images and filtering



|     |                                                 |                                                                                |
|-----|-------------------------------------------------|--------------------------------------------------------------------------------|
| (a) | cp(F) sortF(f $\leq$ f $^\circ$ ) listf filterp | F( $\leq$ ) $\sqsubseteq$ f $\leq$ f $^\circ$ — mon-str, $\triangleq$ thinlist |
| (b) | cp(F) sortF(F( $\leq$ )) listf filterp          | F(sort( $\leq$ )) listcp $\sqsubseteq$ cp(F) sort(F( $\leq$ )) — (8.3b)        |
| (c) | F(sort( $\leq$ )) listcp listf filterp          |                                                                                |

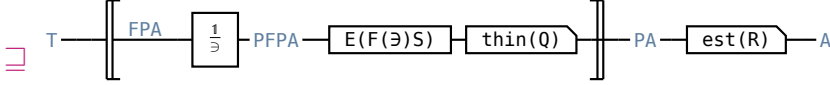
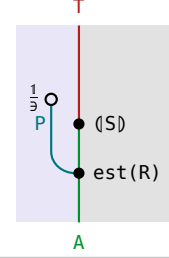
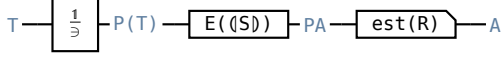
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge \mathbb{1} \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge$   
 $\text{MonoAlg}(f_2 \text{ } p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge$   
 $\text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq$   
 $P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq)$   
 $\wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$   
 $\wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$   
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq)$   
 $\text{thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(S)}{\exists} \text{ est}(R)$

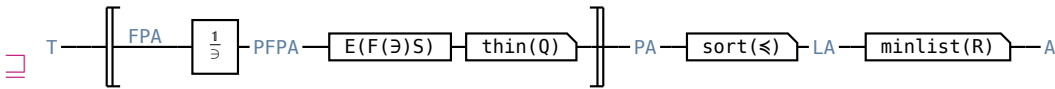
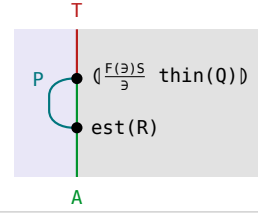
a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

circuit

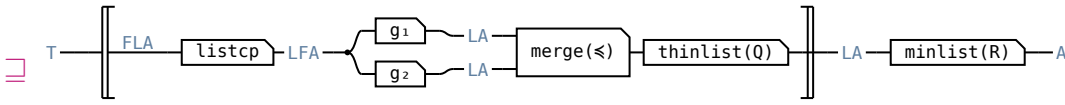
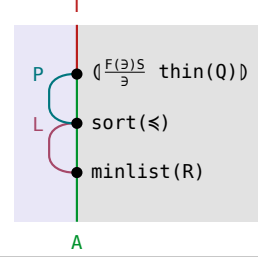
Hinze–Marsden



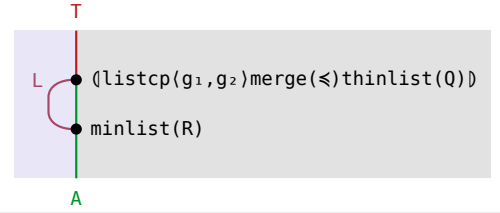
Corollary 8.1 at  $f_1 p_1$  and  $f_2 p_2$  monotonic on  $Q \triangleq \text{thinlist}$



$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$



fusion at (8.3c)



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{ listcp } (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) (f_1 p_1 \cup f_2 p_2)}{\exists} \text{ thin}(Q) \text{ sort}(\leq)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                    | Hinze–Marsden |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| $\sqsubseteq$ <p><math>\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \text{ — (8.3b)}</math></p>                                                                                                                                                                                                                                                                                                     |               |
| $=$ <p><math>\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} \text{ cup sort}(\leq) \text{ thinlist}(Q)</math><br/> <math>S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}</math></p>                                                                                                                                                                            |               |
| $\sqsubseteq$ <p><math>\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q)</math><br/> <math>(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq) \text{ — (8.3b)}</math></p>                                                                                                    |               |
| $=$ <p><math>\{ \frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(\leq) \} \text{ merge}(\leq) \text{ thinlist}(Q)</math><br/> <math>\langle X, Y \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) = \langle X \text{ sort}(\leq), Y \text{ sort}(\leq) \rangle \text{ — bdm-prod-laws}</math></p>                                                                                  |               |
| $\sqsubseteq$ <p><math>F(\text{sort}(\leq)) \text{ listcp } \langle g_1, g_2 \rangle \text{ merge}(\leq) \text{ thinlist}(Q)</math><br/>     Lemma 8.1 at <math>f_1, p_1</math> and at <math>f_2, p_2</math>, then <math>X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_i \triangleq \text{list}(f_i)</math><br/> <math>\text{filter}(p_i) \text{ — } \triangleq \text{thinlist}</math></p> |               |

## §8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{ value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{ weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \circ \text{,}$  within  $w$  the coreflexive on  $xs$  with weight  $xs \leq w, 0 \leq w.$

$R \triangleq \text{value} \geq \text{value}^\circ, Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ), P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times A, \text{ listcp} = \text{wrap} + \text{cpr}, g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1],$   
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w), h_2 \triangleq \text{list}(\pi_2).$

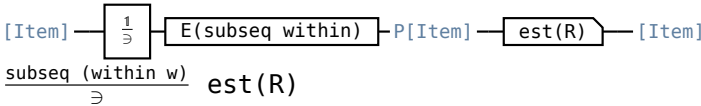
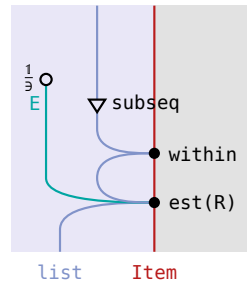
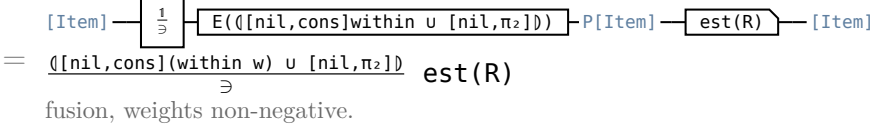
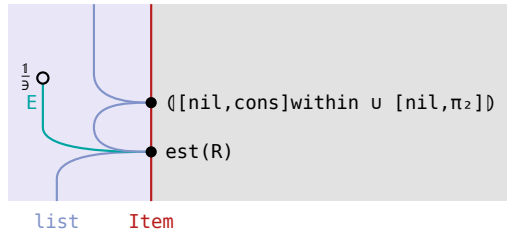
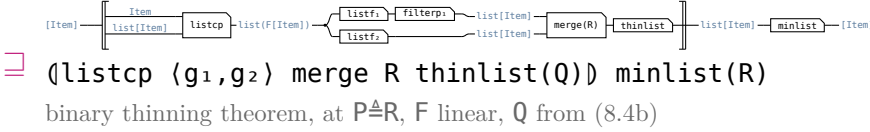
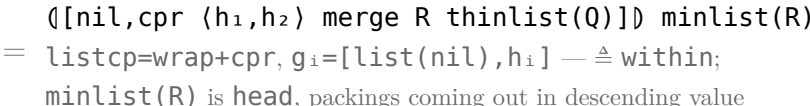
(8.4b)

| the law                                                                                                                                      | what it says                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$<br>FALSE                                      | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on Q once ties in value are broken by weight                         |

(8.4c)

$$\begin{aligned}
& 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R \\
& [\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq \\
& P([\text{nil}, \pi_2]) \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup } \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \\
& \text{sort}(R) \wedge \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow \langle \text{listcp } (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{merge}(R) \text{thinlist} \rangle \\
& \text{minlist} \sqsubseteq \frac{\text{subseq within}}{\supset} \text{est}(R)
\end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                              | Hinze–Marsden                                                                         |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|     |   |
|  |  |
|  |  |
|   |                                                                                       |

## §8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

| the law                                                                                    | what it says                                                                                                             |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| (1×R) glue⊆glue R FALSE                                                                    | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| (1×Q) new⊆new Q<br>(1×Q) (glue (ok w))⊆glue (ok w)Q<br>cons monotonic on collect≤collect°. | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| merge T=cat; P≐head prefix head° also serves                                               | T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| $ \begin{aligned} & (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap} \text{ wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \text{ wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap} \text{ wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \sqsubseteq E(\text{ok}) \text{sort}(\tau) \wedge \\ & F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \text{sort}(\tau) \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ( \text{listcp} (\text{listf}_1, \text{listf}_2 \\ & \text{filterp}_2) \text{merge}(\tau) \text{thinlist} ) \text{minlist} \sqsubseteq \frac{\text{partition fits}}{\supset} \text{est}(R) \end{aligned} $ <p>a paragraph laid out as a fold that thins the layouts kept at each word</p> |                      |
| <b>circuit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Hinze–Marsden</b> |
| $ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits})} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & \frac{\text{partition list}^*(\text{fits } w)}{\supset} \text{est}(R) \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |
| $ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok})} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & = \frac{( [\text{wrap wrap}, \text{new}] \cup (\text{glue } (\text{ok } w)) )}{\supset} \text{est}(R) \end{aligned} $ <p>fusion, every word fits on a line by itself.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                      |
| $ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok})} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & = \frac{( [\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)) )}{\supset} \text{est}(R) \end{aligned} $ <p>the algebra as <math>(f_1 p_1) \cup (f_2 p_2)</math>, <math>p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |
| $ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok})} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & \sqsubseteq ( \text{listcp} (g_1, g_2) \text{cat thinlist}(Q) ) \text{minlist}(R) \end{aligned} $ <p>binary thinning theorem, at <math>P \triangleq T</math> with <math>\text{merge } \tau = \text{cat}</math>, <math>Q</math> from (8.5b)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      |
| $ \begin{aligned} & ( [\text{start}, \text{cpr} (h_1, h_2) \text{cat thinlist}(Q) ] ) \\ & = \text{minlist}(R) \end{aligned} $ <p><math>\text{listcp} = \text{wrap} + \text{cpr}</math>, <math>g_i</math> along the coproduct — <math>\triangleq</math> partition</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      |

## §8.6. Bitonic tours

(8.6a)

$FA = (\text{City} \times \text{City}) + (\text{City} \times A)$ , the base functor of cons-lists of length at least two;  $\text{listcp} = \text{wrap} + \text{cpr}$ ;  $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$ , neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$ ,  $\text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$ ,  $\text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$ ,  $\text{tour} \triangleq ([\text{start}, \text{dropl } \cup \text{dropr}])$ .

$\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys$ ,  $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$ ,  $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$ .

$\text{next} \triangleq \text{tail head}$ ,  $\text{next2} \triangleq \text{next} \times \text{next}$ ,  $\text{head2} \triangleq \text{head} \times \text{head}$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ ,  $Q \triangleq R \cap (\text{next2 } \text{next2}^\circ) \cap (\text{head2 } \text{head2}^\circ)$ ,  $P \triangleq T$ ,  $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$ ,  $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$ .

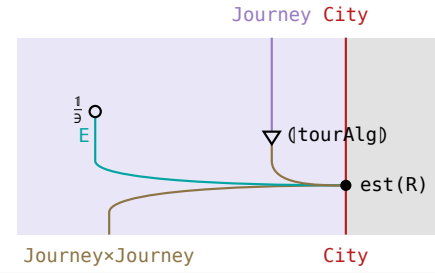
(8.6b)

| the law                                                       | what it says                                                                                                       |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $(1 \times R) \text{dropl} \sqsubseteq \text{dropl } R$ FALSE | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| $(1 \times R) \text{dropr} \sqsubseteq \text{dropr } R$ FALSE |                                                                                                                    |
| $(1 \times Q) \text{dropl} \sqsubseteq \text{dropl } Q$       | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |
| $(1 \times Q) \text{dropr} \sqsubseteq \text{dropr } Q$ ,     |                                                                                                                    |

$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{start}, \text{droplFn}] \tau [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq \quad (8.6c) \\
& P([\text{start}, \text{droplFn}]) \text{sort}(\tau) \wedge \text{sortF}([\text{start}, \text{droprFn}] \tau [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq \\
& P([\text{start}, \text{droprFn}]) \text{sort}(\tau) \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{cat} \\
& \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Qc(tc)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp} \\
& (g_1, g_2) \text{cat thinlist}) \text{minlist} \sqsubseteq \frac{(\text{tourAlg})}{\exists} \text{est}(R) \\
& \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
\end{aligned}$$

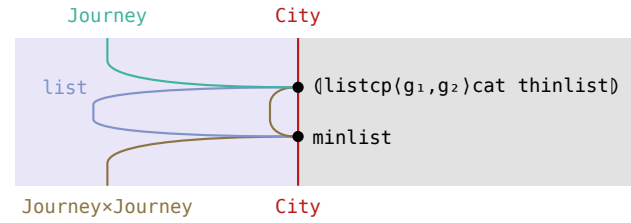
**circuit****Hinze–Marsden**

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} P(\text{Journey}^2) \xrightarrow{\text{est}(R)} \text{Journey} \\
& \frac{\text{tour}}{\exists} \text{est}(R)
\end{aligned}$$



$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} P(\text{Journey}^2) \xrightarrow{\text{est}(R)} \text{Journey} \\
& = \frac{([\text{start}, \text{dropl}] \cup \text{dropr})}{\exists} \text{est}(R) \\
& \text{tour} \triangleq ([\text{start}, \text{dropl}] \cup \text{dropr}) \triangleq \text{tour}
\end{aligned}$$

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{F(\text{list}(\text{Journey}^2))} \text{listcp} \xrightarrow{\text{list}(F(\text{Journey}^2))} \begin{matrix} g_1 \\ g_2 \end{matrix} \xrightarrow{\text{cat}} \text{thinlist} \xrightarrow{\text{list}(\text{Journey}^2)} \text{minlist} \xrightarrow{\text{Journey}} \text{Journey} \\
& \sqsubseteq (\text{listcp} (g_1, g_2) \text{cat thinlist}(Q)) \text{minlist}(R) \\
& \text{binary thinning theorem, at } P \triangleq T \text{ with merge } T = \text{cat}, Q \text{ from (8.6b)}
\end{aligned}$$



$$\begin{aligned}
& ([\text{start wrap}, \text{cpr} (\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\
& = \text{thinlist}(Q)] \text{minlist}(R) \\
& \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\
& \text{quadratic, two tours added per step}
\end{aligned}$$